Mandatory Payment Reform and the Geography of Dialysis Supply: Experimental Evidence from Medicare's End-Stage Renal Disease Treatment Choices Model

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Abstract

Provider payment reforms can affect not only treatment choices but also the availability of care. We study this supply-side margin in Medicare's End-Stage Renal Disease Treatment Choices (ETC) model, which randomly assigned 30 percent of hospital referral regions to mandatory payment adjustments designed to increase home dialysis and transplantation. Using nine archived Centers for Medicare & Medicaid Services (CMS) facility vintages, a fixed 2019 demand grid, and a fixed road network, we estimate the causal effect of ETC assignment on geographic access to in-center dialysis. Our primary outcome is the share of chronic kidney disease (CKD)-risk-weighted population whose second-nearest in-center facility lies more than 30 minutes away, capturing local redundancy in access. At the November 2025 endpoint, the adjusted treated-control contrast is -0.64 percentage points (standard error 0.95), with a 90 percent confidence interval of $[-2.20, +0.92]$. The interval lies within a ± 5-point materiality band, ruling out policy-relevant average deterioration in geographic redundancy. Results are robust to routing uncertainty, alternative estimators, weighting schemes, and travel-time thresholds. We likewise find no familywise-adjusted effects on facility closure, capacity, travel time, or service availability. These findings indicate that, despite meaningful downside payment risk, ETC did not materially reorganize the average local supply of dialysis care during its operating period. Combined with prior evidence of limited patient-side responses, the results suggest that the model had little average effect on either utilization or provider-network margins.

JEL classification: I11, I18, H51, C93.

Keywords: dialysis; Medicare payment reform; provider supply; geographic access; randomized policy experiment; equivalence testing.

I. Introduction

Medicare payment reforms are usually evaluated through the clinical choices they target. Providers can also respond by changing the supply of care. This margin is consequential in dialysis because in-center hemodialysis typically requires three visits each week (Salerno et al., 2022). Longer travel to dialysis is associated with lower quality of life and higher mortality (Moist et al., 2008). Greater distance from a treating nephrologist is also associated with higher mortality (Tonelli et al., 2007). Facility closure is associated with higher hospitalization among displaced patients (Niu et al., 2021) and longer travel to replacement care (Niu et al., 2023). Rural patients face especially long travel to replacement facilities (Stephens et al., 2013). A payment model can therefore leave its intended utilization outcomes largely unchanged while still affecting patients through facility exit or capacity reductions.

The ESRD Treatment Choices (ETC) model offers unusual leverage on this supply question. Beginning in January 2021, CMS required every eligible dialysis facility and managing clinician in 30 percent of hospital referral regions (HRRs) selected by lottery to participate in a payment model (Centers for Medicare & Medicaid Services, 2020b). The incentives rewarded home dialysis and transplantation performance. The first-year randomized evaluation found no detectable increase in home dialysis (Ji et al., 2022). Subsequent studies using data through 2024 found little change in home dialysis and transplantation (Koukounas et al., 2024a; Dixit et al., 2026), apart from a modest early increase in home-dialysis initiation among incident patients (Johansen et al., 2023). CMS ended the model on December 31, 2025, 18 months before its scheduled completion (Centers for Medicare & Medicaid Services, 2025).

The patient-side evidence leaves the provider network unmeasured. That gap matters because public comments on the rule raised the possibility that mandatory downside risk would close facilities or reduce access (Centers for Medicare & Medicaid Services, 2020b, p. 61325). The broader dialysis literature supports the plausibility of this concern. Providers adjust coding, treatment, inputs, and capacity when administered prices change (Dafny,

2005; Acemoglu and Finkelstein, 2008; Clemens and Gottlieb, 2014; Clemens et al., 2021; Lin et al., 2017). Large chains can also reorganize activity after acquisitions (Eliason et al., 2020). Competition shapes provider behavior and outcomes in health care, including dialysis (Gaynor et al., 2015; Erickson et al., 2018). Observational evidence on the 2011 prospective payment bundle finds a lower average probability of closure, though small or hospital-based facilities faced somewhat greater risk (Norouzi et al., 2020). More recently, closures exceeded openings from 2022 onward and were concentrated among smaller facilities and rural communities (Varkila et al., 2026). Those national changes overlap the ETC period. The CMS lottery separates the payment model from the common trend.

We use the lottery to estimate the effect of ETC assignment on the physical configuration of dialysis supply and the potential geographic access that configuration provides. The data combine nine archived CMS facility vintages from 2019 through July 2026 with a fixed 2019 road network. We hold the demand grid and population weights at their baseline values. This construction turns the access outcome into a supply-side accounting measure: after baseline, it can move only when the set of open physical sites changes.

Our primary outcome is the two-site deficit share. It measures the fraction of CKD-risk-weighted population in an HRR whose second-nearest open in-center site lies more than 30 road minutes away. The second site captures geographic redundancy. A patient population with two reachable facilities retains an alternative after one site exits; a population served by a single reachable facility does not. The outcome measures potential access from the observed facility network (Guagliardo, 2004; Luo and Wang, 2003). It does not measure realized patient travel or facility choice (Salerno et al., 2022).

The average effect is small and estimated precisely enough for a policy-relevant equivalence test. At the November 2025 endpoint, six weeks before termination, the baseline-adjusted contrast is -0.64 percentage points with a design-based standard error of 0.95. The 90 percent confidence interval, $[-2.20, +0.92]$, lies inside a ± 5-point materiality band. A five-point change corresponds to roughly 350,000 units of national CKD-risk mass, so the benchmark represents

a policy-scale change in network resilience rather than a negligible numerical deviation. Worst-case treatment-contrast bounds for unroutable origins remain inside the band. The same classification holds for conventional regression adjustments, the CKD-risk-mass-weighted estimand, and every travel-time threshold from 10 to 60 minutes. Inverting the randomization test supports equivalence down to a ± 2.5-point band.

A broader set of outcomes asks whether the threshold measure misses a different form of reorganization. We examine continuous travel times, site closure, station capacity, and the availability of home-dialysis services. Across 31 reported margin cells, the smallest unadjusted randomization p-value is 0.25. No estimate survives its familywise adjustment. The placebo-period contrast is close to zero, and the outcome trajectory remains within the materiality band throughout the model period. Agreement across these outcomes matters because a stable threshold share could otherwise conceal offsetting changes in site counts, capacity, or service mix.

Two exploratory analyses identify narrower patterns. HRRs that relied more heavily on facilities in treated neighboring regions lost more redundancy, conditional on their own assignment. A baseline proxy for expected payment pressure also has a positive treatment interaction. Both magnitudes remain below the materiality band, and the proxy result is concentrated in one component. The estimates rely on additional modeling choices and face a wider multiple-testing problem, so they are best read as directions for evaluations with facility-level payment and patient-flow data.

The main contribution is randomized evidence on provider supply under mandatory payment risk. Existing ETC studies measure patient treatment choices, while the closure literature documents network changes and their consequences. This paper links the two by estimating how the same randomized policy changed physical sites and geographic resilience. The analysis also contributes a reusable access measure built on a fixed road network. Its equivalence framework converts a precise near-zero estimate into an explicit bound on policy-relevant deterioration, which is more informative for policy than reporting statistical

insignificance alone. The paper also states the additional assumptions used for inference when dialysis service markets cross administrative borders.

The paper proceeds from the payment model and its supply mechanism to the data construction. It then presents the research design, the main results and robustness analyses, the exploratory patterns, and the policy interpretation.

II. Institutional Setting and the Supply-Side Question

A. THE ETC PAYMENT MODEL

End-stage renal disease is a disease-specific Medicare entitlement. Most patients without a transplant receive dialysis (Johansen et al., 2024), and two national chains operate a large share of facilities (Eliason et al., 2020). CMS introduced ETC under the Advancing American Kidney Health initiative. The agency selected 30 percent of HRRs through a lottery stratified by the four census regions. Every eligible facility and managing clinician located in a selected HRR participated (Centers for Medicare & Medicaid Services, 2020b). HRRs with at least 20 percent of their ZIP codes in Maryland were selected outside the lottery and are excluded from the randomized analysis.

ETC used two payment adjustments. The home-dialysis payment adjustment increased reimbursement for home-dialysis claims by 3 percent in 2021, 2 percent in 2022, and 1 percent in 2023. Beginning in July 2022, the performance payment adjustment applied to all dialysis claims. It rewarded or penalized home-dialysis and transplant performance. Scheduled facility adjustments eventually reached a maximum bonus of 8 percent and a maximum penalty of 10 percent. Clinician adjustments reached a maximum bonus of 8 percent and a maximum penalty of 9 percent. The model was scheduled to operate through June 2027.

Realized downside exposure increased during implementation. The share of participating facilities receiving a penalty rose from 13.8 percent in 2021 to 25.1 percent in 2023 (Dixit et al., 2026). Facilities serving populations with greater social risk were more likely to be

penalized during the first model year (Koukounas et al., 2024b). CMS announced an early end in March 2025 and finalized termination effective December 31, 2025. The agency cited the absence of improvement in the model's key measures and the absence of expenditure savings (Centers for Medicare & Medicaid Services, 2025).

B. HOW PAYMENT RISK CAN CHANGE GEOGRAPHIC ACCESS

The supply mechanism begins with facility margins. A treated facility that performs below its benchmark faces lower reimbursement on Medicare dialysis claims. Fixed operating costs make that reduction more consequential for low-volume sites. Facilities can respond at the extensive margin through exit. They can also reduce stations or alter the services offered at an existing site. Multi-market ownership creates a further possibility: a chain can reallocate capacity across HRR borders while preserving its national site count. These responses leave different empirical signatures. Exit changes site counts and geographic redundancy, station reductions change capacity, and service reconfiguration changes reported home-dialysis capability. The empirical design follows each margin so that a stable headline access measure cannot conceal an offsetting supply response.

Geography converts these provider decisions into patient exposure. In-center hemodialysis requires frequent attendance, so a modest increase in travel can accumulate into a large time burden. Longer dialysis travel is associated with lower quality of life and higher mortality (Moist et al., 2008). Greater distance from a treating nephrologist is also associated with higher mortality (Tonelli et al., 2007). Patients displaced by a closure experience hospitalization rates 7 to 9 percent above those of matched patients at facilities that remain open (Niu et al., 2021).

The second-nearest site is the relevant resilience margin for this setting. Distance to the nearest site describes current proximity. Distance to the second site describes the alternative available after a single closure (Stephens et al., 2013). The distinction is especially important in thin markets, where losing one facility can change a region from redundant access to a

local monopoly in physical supply.

Prior work cannot isolate the ETC contribution to this network. The 2011 dialysis bundle coincided with a lower average probability of closure, with limited increases among some small or hospital-based units (Norouzi et al., 2020). The later national closure wave began during ETC implementation (Varkila et al., 2026). Patient-level ETC evaluations measure patient outcomes; the facility network remains outside their scope (Ji et al., 2022; Johansen et al., 2023; Koukounas et al., 2024a; Dixit et al., 2026). The randomized HRR assignment therefore supplies the missing comparison for physical supply and geographic access.

III. Data and Measurement

A. FACILITY VINTAGES AND PHYSICAL SITES

The supply data come from the CMS Dialysis Facility archive (Centers for Medicare & Medicaid Services, 2026). We use nine distinct vintages from July 2019 through July 2026. The sequence includes the October 31, 2019 baseline, an August 4, 2020 placebo vintage, and an October 31, 2020 transition vintage. Annual post-implementation vintages cover 2022 through 2025, with November 20, 2025 as the endpoint. The final vintage is July 15, 2026, approximately six and a half months after termination.[1] Each file records the facility certification number (CCN), address, in-center hemodialysis status, station count, chain affiliation, and home-dialysis service indicators.

The physical site is the unit of supply. A facility row enters when it has a valid six-digit CCN, lies in the 50 states or the District of Columbia, and reports in-center hemodialysis. We geocode eligible addresses with the Census Bureau batch geocoder under a fixed cleaning rule (U.S. Census Bureau, 2023). Exact matches form the measurement universe, which covers 91.25 percent of eligible rows. The match rate is stable across vintages and treatment arms.

[1]The archive labeled April 12, 2021 contains a facility file identical to the October 31, 2020 file and is excluded. The facility files inside the October 31, 2019 and August 4, 2020 archives are internally dated December 16, 2019 and August 22, 2020. Both dates precede the September 2020 final rule. Vintage labels refer to archive posting dates.

We collapse facility rows to sites because ownership changes and reorganizations can recycle CCNs. Rows at the same canonical address form one site. Rows within 100 meters also form one site when at least two continuity fields agree. Pairs between 100 and 250 meters are adjudicated without treatment information. The ledger contains 6,413 eligible in-center sites at baseline, 6,355 at the 2025 endpoint, and 6,292 in July 2026. Between baseline and endpoint, 872 baseline sites disappear and 814 new sites enter. Figure A1 reports the national counts. Site-level churn is 13.6 percent, compared with CCN-level churn of 11.0 percent, which is consistent with physical turnover that a CCN-only panel would miss. This national churn is analytically useful because the network is not mechanically static. The randomized comparison can therefore detect whether treated HRRs experienced systematically different turnover or access changes within a moving market.

B. FIXED DEMAND GRID AND POPULATION WEIGHTS

The demand side remains fixed at baseline values. Each populated 2010 census block group receives a weight equal to its 2014–2018 American Community Survey adult population multiplied by the 2020 CDC PLACES modeled prevalence of diagnosed chronic kidney disease in the corresponding census tract (U.S. Census Bureau, 2012, 2019; Centers for Disease Control and Prevention, 2020; Greenlund et al., 2022). We call the product CKD-risk mass. The weight is a population-risk index, and its units do not equal a count of diagnosed patients.

Weights are available for 216,644 of 216,864 populated block groups. The 220 omitted block groups lack a comparable population estimate. The 302 randomized HRRs contain 7.01 million units of CKD-risk mass. We assign block groups and facilities to Dartmouth Atlas HRRs, the unit of randomization (Wennberg and Cooper, 1996). We classify block groups with 2010 USDA rural-urban commuting area codes from 4 through 10 as rural (U.S. Department of Agriculture, Economic Research Service, 2019).

Holding the origins and their weights fixed serves an accounting purpose. Population migration and changes in CKD prevalence cannot move the outcome after baseline. Only

changes in the measured facility configuration can do so. The resulting contrast therefore isolates reorganization of potential supply access rather than changes in where the at-risk population happens to live.

C. ROAD TRAVEL AND THE TWO-SITE DEFICIT SHARE

We route every block-group origin to candidate open sites on the July 2019 OpenStreetMap road network (OpenStreetMap contributors, 2019). OSRM backend 26.5.0 uses the same graph for every facility vintage (Luxen and Vetter, 2011; Project OSRM, 2026). Let $d_{bt}^{(2)}$ denote the road travel time from block group b to its second-nearest open in-center site in vintage t, and let w_b denote fixed CKD-risk mass. For HRR h and threshold c, the deficit share is

$$Y_{ht}(c) = \frac{\sum_{b \in h} w_b \, \mathbf{1}\left\{ d_{bt}^{(2)} > c \right\}}{\sum_{b \in h} w_b}. \tag{1}$$

The primary outcome sets $c = 30$ minutes. Thirty minutes is a travel-time category boundary in Moist et al. (2008). It also has a direct resilience interpretation: an origin counted as deficient lacks a second in-center option within a travel radius that already represents a substantial recurring burden for thrice-weekly treatment. Section V.D reports the complete curve from 10 to 60 minutes.

Equation (1) measures the share of risk-weighted population without two sites inside the threshold. A higher value indicates less redundancy. Because the road network, origin locations, and weights are fixed, the outcome changes only when the set of open sites changes.

At baseline, the equal-HRR mean deficit share is 28.6 percent, with a cross-HRR standard deviation of 25.4. Figure 1 shows substantial geographic dispersion. Weighting origins nationally yields a deficit share of 18.9 percent and a mean second-site travel time of 19.4 minutes. The gap between the equal-HRR and national weighted means reflects the concentration of high-deficit areas in smaller markets. It also explains why the paper reports both a region-average effect and a population-weighted effect.

The primary routing table is complete for 99.76 percent of national CKD-risk mass. The

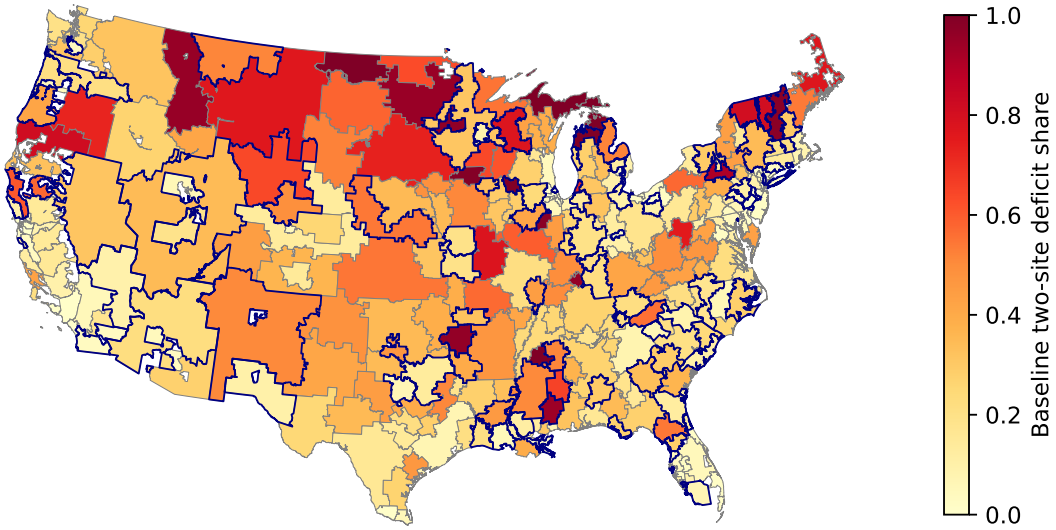


Figure 1: Baseline two-site deficit share by hospital referral region. Shading reports the CKD-risk-weighted share of each HRR whose second-nearest open in-center site lies more than 30 road minutes away at the October 31, 2019 baseline. Dark outlines mark the 91 randomized treated HRRs. Alaska and Hawaii enter the analysis but are omitted from the map.

remaining 0.24 percent consists of 559 block groups with unresolved routes, and the same origin mass is unresolved in every vintage. For each HRR, let D_{ht} be routed deficit mass, M_h unresolved mass, and W_h total mass. The regional outcome lies between

$$Y_{ht}^{L}(c) = \frac{D_{ht}}{W_h} \qquad \text{and} \qquad Y_{ht}^{U}(c) = \frac{D_{ht} + M_h}{W_h}. \tag{2}$$

The lower value assigns every unresolved origin to the non-deficit state. The upper value assigns every unresolved origin to the deficit state. Section V.A combines these regional bounds in the direction that minimizes or maximizes the treated-control contrast.

The outcome also changes enough over time to detect network movement. From baseline to the 2025 endpoint, the standard deviation of HRR-level changes is 8.5 percentage points. A total of 119 HRRs move by more than one point, and 48 move by more than five points. The corresponding standard deviation during the 2019 to August 2020 placebo interval is 2.5 points. Figure 2 reports both the outcome changes and the baseline distribution of

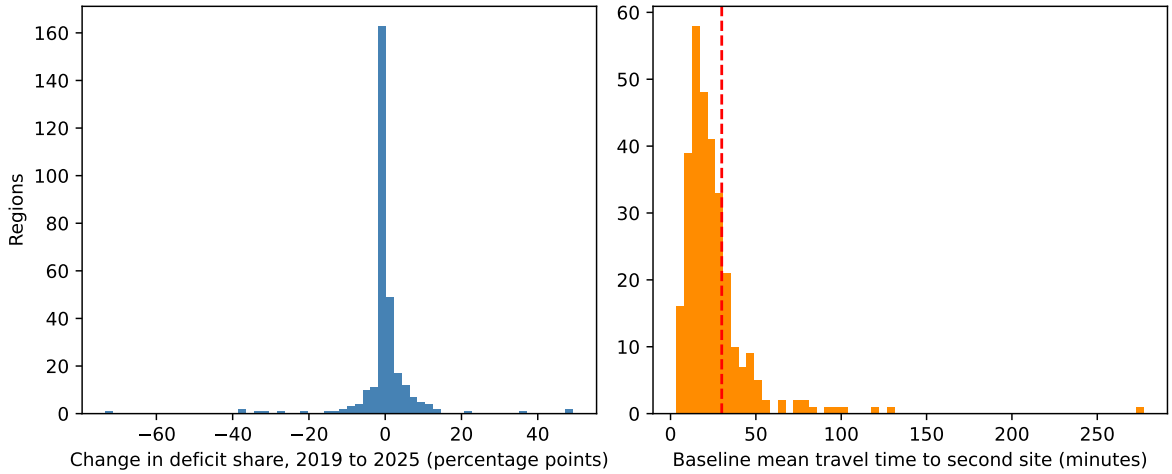


Figure 2: Movement in the primary outcome and baseline distance to the threshold. The left panel shows the distribution across 302 HRRs of the change in the two-site deficit share from baseline to the 2025 endpoint. The right panel shows the distribution of baseline mean travel time to the second-nearest site. The dashed line marks 30 minutes.

second-site travel time. Nationally, 8.5 percent of CKD-risk mass lies within five minutes of the 30-minute threshold, while 2.1 percent crosses that threshold between baseline and endpoint. These movements show that the outcome is responsive to actual network changes; the main result is therefore a treatment-control null in a moving market, rather than a null created by an inert measure.

D. RANDOMIZED ASSIGNMENT AND BASELINE BALANCE

Treatment status comes from the CMS selected-HRR report (Centers for Medicare & Medicaid Services, 2020a). We reconstruct assignment from the published spreadsheet and cross-check the result against the report. CMS listed 95 selected HRRs. Four entered under the Maryland provision and fall outside the lottery. The analysis therefore contains 91 treated HRRs and 211 controls.

The exact randomization strata are the four census regions. Treated counts are 25 of 84 HRRs in the Midwest, 13 of 43 in the Northeast, 35 of 115 in the South, and 18 of 60 in the West. Table 1 reports baseline means, standardized differences, and randomization p-values. The treated group has a baseline deficit share 3.0 points above the control group

Table 1: Baseline balance across randomized hospital referral regions

Baseline variable	Treated	Control	Difference	Std. difference	Randomization p
Two-site deficit share, 30 minutes (percent)	30.7	27.7	+3.0	+0.12	0.34
Mean travel time to second-nearest site (minutes)	27.8	24.6	+3.1	+0.12	0.27
In-center physical sites	21.1	20.5	+0.6	+0.03	0.85
Dialysis stations	385	381	+4	+0.01	0.94
Share of facilities offering peritoneal dialysis (percent)	52.3	54.0	−1.7	−0.09	0.47
Share offering home hemodialysis training (percent)	32.1	31.4	+0.7	+0.05	0.73
Rural share of CKD-risk mass (percent)	28.0	25.4	+2.6	+0.11	0.38
CKD-risk mass (thousands)	23.6	23.0	+0.6	+0.02	0.85

Notes: Means cover 91 treated and 211 control HRRs at the October 31, 2019 baseline. Standardized differences divide the treated-control difference by the pooled standard deviation. Randomization p-values use 20,000 draws from the exact census-region-stratified assignment mechanism. CKD-risk mass is the fixed weight defined in Section III.B.

and a second-site travel time 3.1 minutes longer. Each standardized difference is 0.12, and the corresponding randomization p-values are 0.34 and 0.27. Other baseline differences appear in prior descriptions of the randomized facilities (Wilk et al., 2022). The primary estimator uses the baseline value of each outcome to improve precision and absorb this chance imbalance. Economically, the adjustment compares HRRs around their own starting access conditions, which prevents the treated group's slightly thinner baseline network from being mistaken for a policy effect.

IV. Research Design

A. ESTIMANDS

The CMS lottery identifies an intention-to-treat contrast for regional assignment. Let $\mathbf{Z}$ denote the full assignment vector and let $Y_i(\mathbf{Z})$ denote the endpoint outcome for HRR i. Allowing an HRR's outcome to depend on assignments elsewhere, the design-specific own-assignment estimand is

$$\tau = \frac{1}{N}\sum_{i=1}^{N}\left[\mathbb{E}\{Y_i(\mathbf{Z}) \mid Z_i = 1\} - \mathbb{E}\{Y_i(\mathbf{Z}) \mid Z_i = 0\}\right], \tag{3}$$

where the expectations use the realized stratified assignment mechanism. This estimand averages HRRs equally and describes the effect of own assignment under the exposure distribution induced by the CMS lottery. It remains tied to that assignment mechanism when cross-border interference is present.

A second estimand weights each HRR by its CKD-risk mass. That contrast describes the average assignment effect for the measured at-risk population. The equal-HRR estimand describes the average randomized region. We report both policy objects because large urban HRRs contain more CKD-risk mass and tend to have more precise access measures. Agreement between them would show that the conclusion is not driven either by many small markets or by a few population-dense HRRs.

B. BASELINE-ADJUSTED ESTIMATOR

Access outcomes are highly persistent. Baseline adjustment can therefore recover much of the precision available in a 302-HRR experiment. For the observed outcome, write Z_i for assignment, π_i for the treatment probability in HRR i's census-region stratum, Y_i for the endpoint value, and X_i for the baseline value. Within arm $a \in \{0, 1\}$, we estimate a common baseline slope and stratum-specific arm means:

$$m_a(i) = \bar{Y}_{a,r(i)} + \widehat{\beta}_a \left(X_i - \bar{X}_{a,r(i)} \right), \tag{4}$$

where $r(i)$ indexes the census-region stratum. The primary generalized regression, or augmented inverse-probability-weighted, estimator is

$$\widehat{\tau} = \frac{1}{N} \sum_{i=1}^{N} \left[m_1(i) - m_0(i) + \frac{Z_i\{Y_i - m_1(i)\}}{\pi_i} - \frac{(1 - Z_i)\{Y_i - m_0(i)\}}{1 - \pi_i} \right]. \tag{5}$$

Predictions are clipped to the unit interval for share outcomes. They remain unclipped for outcomes measured in minutes, log points, or per-capita units. Equation (5) is a design-based regression adjustment with arm-specific slopes in the spirit of Lin (2013), which addresses the

regression-adjustment concerns in Freedman (2008). Baseline outcomes enter only to improve prediction within the randomized design; treatment remains the CMS lottery assignment, and the estimand remains an intention-to-treat contrast.

The baseline covariate carries most of the available precision. The unadjusted stratified contrast has a standard error of 3.26 percentage points for the primary outcome. Adjustment reduces it to 0.95, which corresponds to a variance reduction of 91.6 percent. The resulting minimum detectable effect at 80 percent power is approximately 2.65 points, compared with 9.14 points for the unadjusted contrast. This gain is central to the paper's interpretation because it allows the experiment to distinguish a policy-relevant small effect from an imprecise failure to reject zero. Appendix Table A2 reports the precision curve.

C. INFERENCE AND CROSS-REGION INTERFERENCE

The main standard error is the stratified Neyman variance estimator applied to the HRR contributions in equation (5) (Neyman, 1990). Under the conventional no-interference potential-outcomes model, this variance is conservative in expectation because the cross-arm covariance of potential outcomes is unidentified. Since dialysis markets cross HRR boundaries, we distinguish that Wald construction from the randomization procedures used elsewhere in the paper (Aronow and Samii, 2017; Basse and Airoldi, 2018; Puelz et al., 2022). The Wald interval summarizes uncertainty around the average regional contrast, while the randomization procedures ask how unusual the observed contrast is under assignments generated by the CMS lottery.

Every headline estimate also receives a Fisher randomization p-value (Fisher, 1935; Imbens and Rubin, 2015). We redraw assignment within the four census-region strata while preserving the realized treated count in each stratum. The test statistic is studentized and recomputed after each draw (Zhao and Ding, 2021). For B simulated assignments, the reported two-sided value is

$$p_{\mathrm{MC}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{|T^{(b)}| \geq |T^{\mathrm{obs}}|\right\}}{B + 1}. \tag{6}$$

Primary cells use 100,000 draws. Margin cells use 20,000, while the simultaneous threshold band uses 5,000 joint draws. A reported value of 0.00001 is the simulation floor when no draw is at least as extreme as the observed statistic. The reported simulation diagnostics include draw counts, exceedance counts, and Monte Carlo intervals.

The global Fisher test imposes the sharp null that every HRR outcome is unchanged under every assignment vector. That null can be tested without separately restricting the form of interference. The Wald interval uses the conventional no-interference variance model. The randomization-based equivalence test in Section IV.D adds a constant-additive-effect model. The cross-border analysis in Section VI.A examines one observed exposure pattern and cannot identify every form of interference. Reading the procedures together separates three questions: whether any assignment effect is detectable, whether the average effect is bounded within a policy-relevant range, and whether effects vary with measured cross-border exposure.

D. EQUIVALENCE AND THE MATERIALITY BAND

A conventional failure to reject zero does not determine whether an estimated effect is small enough to disregard. We evaluate the two-site deficit share against a symmetric materiality band of ± 5 percentage points. The decision rule has three outcomes. We classify a contrast as materially positive or negative when its 95 percent confidence interval lies entirely beyond one edge of the band. We classify it as equivalent when its 90 percent interval lies entirely inside the band, which implements the two one-sided tests procedure at the 5 percent level (Schuirmann, 1987). All other intervals are indeterminate.

The five-point benchmark has a direct scale interpretation. A change of that size corresponds to about 350,000 units of national CKD-risk mass, approximately one quarter of the baseline national deficit and one fifth of the cross-HRR standard deviation. It therefore marks a change large enough to alter the policy assessment of network resilience. The adjusted design's 80 percent-power minimum detectable effect is about 2.65 points, which is smaller

than the band and makes the equivalence exercise empirically informative.

We also invert a randomization test for each band edge. Under the constant-additive model $Y_i(z) = Y_i(0) + \tau z_i$, the hypothesis $\tau = \tau_0$ is sharp after subtracting $\tau_0 Z_i$ from each observed outcome. Re-randomization of the effect-removed outcomes tests each edge without an estimated variance. Inverting the two tests over a grid gives the equivalence frontier, defined as the smallest symmetric band supported by this procedure.

E. OUTCOME FAMILIES AND REPORTING STRUCTURE

The analysis centers on the 30-minute two-site deficit share because it translates facility configuration into a direct measure of resilience to a single local closure. The results then widen in a conceptual sequence: routing bounds, alternative estimators and weights, the outcome trajectory, the full threshold curve, continuous travel times, physical capacity, and reported service availability. This sequence asks whether the main conclusion survives both measurement uncertainty and economically distinct forms of provider response.

The cross-border exposure and payment-pressure sections are presented as exploratory evidence. Their interpretation is intentionally cautious because they depend on constructed moderators, additional functional-form choices, and a larger multiple-testing problem. Every table reports the uncertainty needed to compare the magnitude of those patterns with the average effect and with the five-point materiality benchmark.

V. Results

A. PRIMARY EFFECT ON TWO-SITE RESILIENCE

Table 2 reports the endpoint result. The adjusted treated-control contrast is -0.64 percentage points, with a stratified Neyman standard error of 0.95. A negative estimate indicates slightly less deficit in treated HRRs. The 95 percent confidence interval is $[-2.50, +1.21]$, and the two-sided randomization p-value is 0.52. Its Monte Carlo 95 percent interval is 0.517 to

Table 2: Effect of ETC assignment on the two-site deficit share

Outcome	Estimate	SE	90% CI	95% CI	p
Two-site deficit share, 30 minutes, 2025 endpoint	−0.64	0.95	[−2.20, +0.92]	[−2.50, +1.21]	0.52
unadjusted stratified difference	+2.23	3.26	[−3.13, +7.59]	[−4.16, +8.62]	0.50
lower bound (missing mass against deterioration)	−1.02	0.92	[−2.53, +0.50]	[−2.82, +0.79]	
upper bound (missing mass toward deterioration)	−0.21	0.94	[−1.75, +1.34]	[−2.05, +1.63]	
Pre-policy placebo: change 2019 to August 2020	+0.02	0.25	[−0.39, +0.43]	[−0.46, +0.50]	0.96
Two-site deficit share, 60 minutes, 2025 endpoint	−0.26	0.79	[−1.56, +1.03]	[−1.81, +1.28]	
lower bound	−0.60	0.76	[−1.86, +0.65]	[−2.10, +0.89]	
upper bound	+0.20	0.77	[−1.07, +1.47]	[−1.32, +1.72]	

Notes: Estimates are in percentage points and equal treated minus control across 302 randomized HRRs. Rows without a suffix use equation (5). The unadjusted row is the stratified Horvitz-Thompson difference. Standard errors are stratified Neyman. Two-sided randomization p-values use 100,000 draws from the exact stratified design. For the missing-route bounds, the lower contrast applies Y^L to treated HRRs and Y^U to controls; the upper contrast reverses those assignments. The placebo outcome is the change from October 31, 2019 to August 4, 2020.

0.523. The point estimate modestly favors treated HRRs, while the interval spans zero; the substantive conclusion is a tightly bounded average effect rather than evidence of an access improvement.

The 90 percent interval, $[-2.20, +0.92]$, lies inside the ± 5-point band. The data therefore reject average deterioration of five points and average improvement of the same size at the 5 percent level. The estimate is also substantively small relative to the baseline equal-HRR mean of 28.6 percent. In policy terms, the experiment rules out an average loss of two-site resilience large enough to move roughly one fifth of a typical HRR's baseline deficit scale.

Routing uncertainty does not change this conclusion. Equation (2) gives a lower and upper outcome for every HRR. To minimize the treated-control contrast, we assign the lower outcome to each treated HRR and the upper outcome to each control. This calculation yields -1.02 points, with a 90 percent interval of $[-2.53, +0.50]$. Reversing the assignments maximizes the contrast and yields -0.21 points, with a 90 percent interval of $[-1.75, +1.34]$. Both worst-case contrast intervals remain inside the materiality band.

Baseline adjustment is essential for the strength of the conclusion. The unadjusted stratified difference is $+2.23$ points with a standard error of 3.26. Its 90 percent interval, $[-3.13, +7.59]$, crosses the upper edge of the band. Baseline adjustment changes the point

estimate by less than one unadjusted standard error, but it reduces uncertainty enough to distinguish equivalence from an indeterminate result. The main gain therefore comes from using persistent baseline access to remove noise, rather than from moving the estimated effect toward zero.

The randomization-based TOST reaches the same classification. Neither band-edge null is approached in 100,000 draws, which places both one-sided values at the Monte Carlo floor. Inverting those tests on a 0.25-point grid yields an equivalence frontier of ± 2.50 points. Replacing the Neyman standard error of 0.0095 with the sharp-null randomization dispersion of 0.0106 gives a 90 percent sensitivity interval of $[-2.4, +1.1]$. That interval remains inside the five-point band. Agreement between the Wald and randomization calculations shows that the bounded conclusion does not depend on a single variance formula.

B. ALTERNATIVE ESTIMATORS AND POLICY WEIGHTS

Table 3 compares the primary estimator with conventional baseline-adjusted alternatives. ANCOVA with a common slope gives -0.60 points. A regression with arm-specific slopes gives -0.65, and the change score gives -0.82. Combining census-region-specific regressions yields -0.87. Their standard errors range from 0.84 to 0.97, and every 90 percent interval lies inside the band. The narrow range of estimates indicates that the conclusion is not driven by the functional form of the adjustment.

The CKD-risk-mass-weighted regression estimates the effect for the average measured at-risk person. Its point estimate is -0.54 points, with a standard error of 0.39 and a 90 percent interval of $[-1.19, +0.11]$. The gain in precision reflects the lower weight placed on small HRRs with highly variable access outcomes. Leave-one-HRR-out estimates range from -1.05 to -0.35 points. Excluding the two HRRs with the lowest routing coverage gives -0.43 points with a standard error of 0.94. The equivalence classification is stable across these choices. The population-weighted result also shows that a near-zero region average is not concealing deterioration in the places where most measured CKD-risk mass resides.

Table 3: Alternative estimators and estimands for the endpoint contrast

Specification	Estimate	SE	90% CI	Within band
GREG/AIPW contrast (primary)	−0.64	0.95	[−2.20, +0.92]	yes
ANCOVA, common slope	−0.60	0.97	[−2.20, +1.00]	yes
Regression with arm-specific slope (Lin)	−0.65	0.92	[−2.16, +0.87]	yes
Change score	−0.82	0.95	[−2.39, +0.75]	yes
Region-specific Lin regressions, combined	−0.87	0.84	[−2.25, +0.50]	yes
CKD-mass-weighted Lin regression	−0.54	0.39	[−1.19, +0.11]	yes
Unadjusted stratified difference	+2.23	3.26	[−3.13, +7.59]	no
Leave-one-region-out range of the primary (302 fits)			[−1.05, −0.35]	yes
Primary without the two lowest-coverage regions	−0.43	0.94	[−1.98, +1.11]	yes
Midwest (n = 84)	−0.57	1.73	[−3.42, +2.28]	yes
Northeast (n = 43)	−1.49	1.66	[−4.23, +1.24]	yes
South (n = 115)	−1.30	0.96	[−2.88, +0.28]	yes
West (n = 60)	−0.02	2.65	[−4.38, +4.33]	yes

Notes: Estimates are in percentage points. The GREG/AIPW row uses the primary estimator and stratified Neyman standard errors. Regression rows use HC2 standard errors. The CKD-risk-mass-weighted row uses weighted least squares with regional mass as the weight. Census-region-specific regressions are combined with weights proportional to the number of HRRs in each stratum. "Within band" means that the 90 percent confidence interval lies inside the ±5-point materiality band.

The four census-region contrasts are individually imprecise. Their 90 percent intervals range from $[-4.38, +4.33]$ in the West to $[-2.88, +0.28]$ in the South. These estimates permit regional differences of several points even though the national average is equivalent within five points. Section V.F reports additional heterogeneity and extreme-access checks.

C. PLACEBO-PERIOD CHECK AND OUTCOME TRAJECTORY

The placebo-period check uses the change from the 2019 baseline to the August 2020 vintage, which predates the final rule and the release of selected HRRs. The adjusted contrast is +0.02 points with a standard error of 0.25 and a randomization p-value of 0.96. The unadjusted contrast is +0.003 points. These estimates show that the access measure did not exhibit meaningful treated-control drift during the earlier interval, which supports the comparability of the subsequent trajectory.

Figure 3 traces the adjusted contrast across every usable post-baseline vintage. The estimates are +0.03 points in October 2020, −0.05 in November 2022, −0.10 in October 2023,

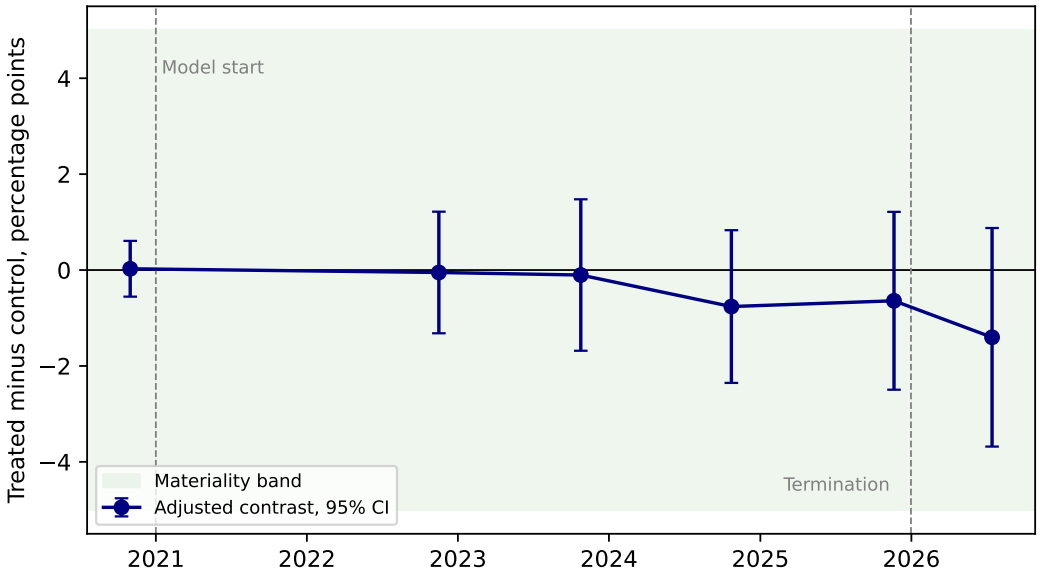


Figure 3: Adjusted treated-control contrast in the two-site deficit share by facility vintage. Points show estimates in percentage points with 95 percent confidence intervals. The shaded area is the ±5-point materiality band. Vertical dashed lines mark model implementation in January 2021 and termination in December 2025.

and −0.76 in October 2024. The November 2025 endpoint is −0.64, followed by −1.40 in July 2026. Standard errors increase from 0.30 in the transition vintage to 1.16 in July 2026. Every point estimate and confidence interval remains well inside the five-point band. The absence of a discrete break implies gradual, modest movement rather than a large reorganization tied to a single implementation year.

The modest late-sample movement is toward lower deficit in treated HRRs. It remains small relative to the five-point band, and the post-termination estimate is less precise. The trajectory therefore supports the endpoint interpretation without establishing a distinct delayed response.

D. TRAVEL-TIME THRESHOLDS

The 30-minute threshold is a convention, so we reconstruct the outcome at every integer cutoff from 10 through 60 minutes. For each cutoff, the same-threshold baseline outcome enters as the covariate. Figure 4 shows a smooth curve close to zero. Point estimates range from

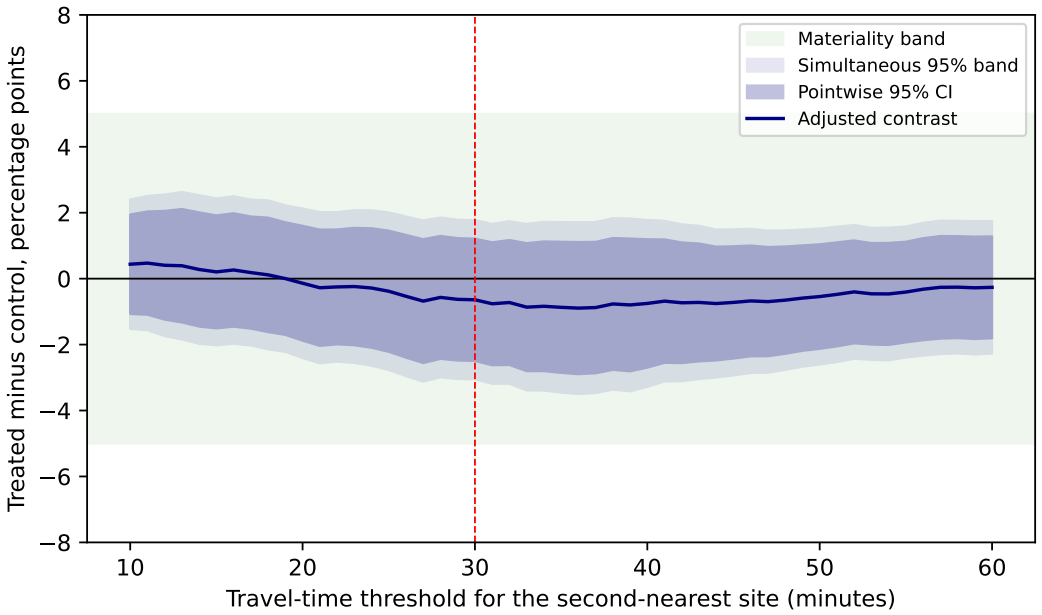


Figure 4: Endpoint contrast across second-site travel-time thresholds. The line reports the adjusted estimate in percentage points at each integer threshold from 10 through 60 minutes. The darker interval is the pointwise 95 percent confidence band. The lighter interval is the simultaneous 95 percent sup-t band. The dashed line marks the 30-minute benchmark used for the headline outcome.

-0.89 to $+0.47$ percentage points. The continuity of the curve matters because it shows that the result is not produced by selecting a cutoff where the contrast happens to be smallest.

Every pointwise 90 percent interval lies inside the five-point band. A simultaneous 95 percent band uses the joint sharp-null randomization distribution of the maximum studentized statistic across all 51 cutoffs. Its critical value is 2.55, and its largest half-width is 2.62 points. At the neighboring cutoffs of 27 and 33 minutes, the estimates are -0.68 and -0.86 points. The 30-minute estimate is -0.64. The joint randomization p-value for the full curve is 0.86. The uniform result implies that ETC did not shift the distribution of second-site access at short, conventional, or extreme travel radii. Appendix Table A1 reports selected cutoffs.

E. ACCESS, CAPACITY, AND SERVICE MARGINS

A threshold share can miss changes that occur away from the cutoff. It can also remain stable when site counts, capacity, or service availability move in offsetting directions. Tables 4

Table 4: Geographic access margins at the 2025 and 2026 vintages

Outcome	Estimate	SE	95% CI	p	Holm p
Second-site travel time, 2025 (minutes)	+0.46	0.89	[−1.29, +2.20]	0.66	1.00
Second-site travel time, 2025 (log points)	−0.6	1.5	[−3.6, +2.4]	0.72	1.00
First-site travel time, 2025 (minutes)	−0.23	0.20	[−0.62, +0.15]	0.28	1.00
First-site travel time, 2025 (log points)	−0.7	0.8	[−2.4, +0.9]	0.39	1.00
Second-site travel time, 2026 (minutes)	+0.04	0.94	[−1.81, +1.88]	0.97	1.00
Second-site travel time, 2026 (log points)	−1.5	1.7	[−4.9, +1.9]	0.39	1.00
Deficit share, 15 minutes, 2025 (pp)	+0.21	0.87	[−1.51, +1.92]	0.82	1.00
Deficit share, 60 minutes, 2025 (pp)	−0.26	0.79	[−1.81, +1.28]	0.77	1.00
Deficit share, 15 minutes, 2026 (pp)	−0.36	1.00	[−2.32, +1.60]	0.73	1.00
Deficit share, 60 minutes, 2026 (pp)	−0.49	0.81	[−2.07, +1.09]	0.58	1.00
Deficit share, 30 minutes, 2026 (pp)	−1.40	1.16	[−3.68, +0.88]	0.25	1.00
Deficit share, 30 minutes, adult weights, 2025 (pp)	−0.43	0.98	[−2.36, +1.49]	0.68	1.00
Second-site time, adult weights, 2025 (minutes)	+0.63	0.95	[−1.24, +2.50]	0.56	1.00
Deficit share, 30 minutes, pooled 2024 to 2026 (pp)	−0.94	0.91	[−2.72, +0.85]	0.33	0.65
Second-site time, pooled 2024 to 2026 (minutes)	+0.16	0.73	[−1.27, +1.58]	0.86	0.86

Notes: Adjusted estimates use stratified Neyman standard errors and 95 percent confidence intervals. Units appear in the row labels; pp denotes percentage points. Randomization p-values use 20,000 stratified draws. Holm p-values adjust within the travel-time, threshold, weighting, and pooled families. Log outcomes are 100 times the log difference. Adult-population weights replace CKD-risk mass.

and 5 report 31 cells across the broader access and supply menu. Holm adjustments are applied within outcome families (Holm, 1979). These measures translate the institutional mechanism into observables: travel times capture proximity, closures and stations capture physical capacity, and service flags capture the model's rewarded home-dialysis margin.

The continuous access outcomes are tightly centered near zero. At the 2025 endpoint, mean travel time to the second-nearest site changes by +0.46 minutes with a standard error of 0.89. Its 95 percent interval is $[-1.29, +2.20]$ minutes, compared with a baseline equal-HRR mean of 25.6. The log specification gives −0.6 log points with a standard error of 1.5. Mean travel time to the nearest site changes by −0.23 minutes with a standard error of 0.20. In July 2026, the second-site estimate is +0.04 minutes. The combination of a near-zero nearest-site effect and a tightly bounded second-site effect indicates stability in both immediate proximity and backup access.

The result is also stable to different population weights and time aggregation. Adult-population weighting gives −0.43 points for the 30-minute deficit share and +0.63 minutes

Table 5: Facility capacity and service margins at the 2025 and 2026 vintages

Outcome	Estimate	SE	95% CI	p	Holm p
Share of baseline sites closed by 2025 (pp)	+0.29	1.56	[−2.76, +3.34]	0.86	1.00
Share of baseline sites closed by 2026 (pp)	+0.01	1.56	[−3.05, +3.08]	0.99	1.00
Sites, 2025 (log points)	−0.5	2.2	[−4.8, +3.8]	0.82	1.00
Sites, 2026 (log points)	+0.4	2.4	[−4.3, +5.1]	0.86	1.00
Stations, 2025 (log points)	−1.5	2.1	[−5.7, +2.6]	0.48	1.00
Stations, 2026 (log points)	−0.4	2.4	[−5.0, +4.3]	0.88	1.00
Stations per 1,000 CKD-risk mass, 2025	−0.24	0.27	[−0.76, +0.29]	0.39	1.00
Stations per 1,000 CKD-risk mass, 2026	−0.12	0.30	[−0.70, +0.47]	0.70	1.00
Share of facilities offering peritoneal dialysis, 2025 (pp)	+1.16	1.32	[−1.42, +3.74]	0.38	1.00
Share offering peritoneal dialysis, 2026 (pp)	+1.14	1.39	[−1.58, +3.86]	0.42	1.00
Share offering home hemodialysis training, 2025 (pp)	+0.02	1.19	[−2.31, +2.35]	0.99	1.00
Share offering home hemodialysis training, 2026 (pp)	−0.43	1.25	[−2.87, +2.02]	0.73	1.00
Share offering a late shift, 2025 (pp)	+0.50	0.84	[−1.14, +2.14]	0.56	1.00
Share offering a late shift, 2026 (pp)	+0.77	0.88	[−0.96, +2.51]	0.39	1.00
Share offering in-center hemodialysis, 2025 (pp)	+0.35	0.43	[−0.50, +1.20]	0.45	1.00
Share offering in-center hemodialysis, 2026 (pp)	+0.28	0.44	[−0.58, +1.13]	0.55	1.00

Notes: Adjusted estimates use stratified Neyman standard errors and 95 percent confidence intervals. Closure shares are the fraction of baseline sites absent at the endpoint. Site and station outcomes are log changes from baseline, reported in log points. Stations per 1,000 units of CKD-risk mass are in levels. Service shares use raw CMS facility rows mapped to HRRs through the site ledger. Randomization p-values use 20,000 stratified draws; Holm p-values adjust within the capacity-and-service family.

for second-site travel. Pooling the 2024 through 2026 vintages gives −0.94 points for the deficit share and +0.16 minutes for second-site travel. None of the threshold cells approaches the materiality band. The pooled estimates also reduce sensitivity to the exact endpoint date and still indicate little sustained divergence between treatment arms.

The facility outcomes show no detectable average retrenchment. The treated-control difference in the share of baseline sites closed by 2025 is +0.29 points with a standard error of 1.56. Nationally, 13.6 percent of baseline sites disappear by that endpoint. The log change in site count is −0.5 points. Station capacity changes by −1.5 log points, with a 95 percent interval of $[-5.7, +2.6]$. Stations per 1,000 units of CKD-risk mass change by −0.24 with a standard error of 0.27. The sector experienced substantial turnover, yet ETC regions did not experience systematically greater closure or capacity loss than controls.

Service availability also remains stable. The share of facilities offering peritoneal dialysis changes by +1.16 points. Home-hemodialysis training changes by +0.02 points, while late-shift availability changes by +0.50. The 2026 estimates are similar. Across Tables 4 and 5,

the smallest unadjusted randomization p-value is 0.25. The smallest displayed Holm-adjusted value is 0.65. The incentives therefore did not translate into visible average changes in the facility service flags available in the public data. These facility-level patterns align with the patient-side evidence showing little change in home-dialysis use (Ji et al., 2022; Koukounas et al., 2024a; Dixit et al., 2026).

F. HETEROGENEITY AND EXTREME-ACCESS CHECKS

Table 6 reports three checks in its upper panel. The rural-share interaction asks whether ETC had larger effects in thin rural markets. The baseline-deficit interaction asks whether initially fragile networks responded differently. The 60-minute outcome isolates extreme second-site travel burden. The estimated rural interaction is -0.047, with an HC2 standard error of 0.141 and a randomization p-value of 0.83. The baseline-deficit interaction is $+0.041$, with a standard error of 0.064 and a p-value of 0.57. The 60-minute average contrast is -0.26 points, with a standard error of 0.79 and a p-value of 0.77. Holm-adjusted values equal 1.00.

The moderation intervals are wide. For example, the rural interaction interval is approximately ± 0.28 per unit of rural share. These checks therefore provide limited information about rural-specific harm even though their point estimates are small. The equivalence conclusion applies to the average HRR and to the population-weighted average; it does not establish equivalence for every subgroup or region.

VI. Exploratory Evidence on the Location of Pressure

The analyses in this section ask where a small average effect may be concentrated. Their randomization p-values describe patterns under the stated specifications, while the constructed moderators and wider testing set call for cautious interpretation. A paper-wide familywise correction across the roughly 40 tests would remove conventional statistical significance from both headline patterns. The relevant question is therefore whether the estimates reveal economically coherent locations of pressure that deserve direct testing with richer data.

Table 6: Heterogeneity, extreme-access, and cross-boundary exposure estimates

Test	Coefficient	SE	p	Holm p
Treatment × rural share of CKD-risk mass	−0.047	0.141	0.83	1.00
Treatment × baseline deficit share	+0.041	0.064	0.57	1.00
Deficit share, 60 minutes, 2025 (pp)	−0.26	0.79	0.77	1.00
Treated-neighbor exposure: deficit share, 30 minutes	+0.201	0.095	0.008	0.02
Treated-neighbor exposure: second-site time (minutes)	+11.409	8.063	0.116	0.12
deficit share, controlling for total cross-boundary reliance	+0.250	0.111	0.008	
deficit share, excluding the lowest-coverage region	+0.222	0.094	0.005	
deficit share, excluding the largest mover	+0.163	0.091	0.024	
deficit share, excluding the three largest movers	+0.159	0.078	0.025	

Notes: The upper panel reports the rural-share interaction, the baseline-deficit interaction, and the 60-minute extreme-access contrast. Interaction coefficients come from arm-interacted endpoint regressions with HC2 standard errors. The 60-minute row uses the primary estimator and a stratified Neyman standard error. Randomization p-values use 100,000 stratified draws, and Holm values adjust across the three checks. The lower panel reports coefficients on centered treated-neighbor exposure from regressions containing own assignment, census-region strata, the baseline outcome, and the assignment-by-baseline interaction. Exposure is recomputed in each of 20,000 randomization draws. The displayed Holm values adjust across the two main exposure outcomes.

A. CROSS-BOUNDARY EXPOSURE

Regional assignment does not align perfectly with dialysis service markets. At baseline, 10.3 percent of the first two access slots for an average HRR are supplied by facilities across an HRR border. Cross-border reliance exceeds 5 percent in 211 of 302 HRRs. We define treated-neighbor exposure as the share of an HRR's baseline first-two access slots supplied by facilities in other HRRs that receive ETC assignment. Its mean is 0.033, its standard deviation is 0.059, and its maximum is 0.62.

The lower panel of Table 6 adds centered treated-neighbor exposure to a regression containing own assignment, census-region strata, the baseline outcome, and the assignment-by-baseline interaction. The exposure coefficient for the 30-minute deficit share is +0.201 with an HC2 standard error of 0.095. The randomization p-value is 0.008, and the Holm-adjusted value across the two main exposure outcomes is 0.02. A one-standard-deviation increase in exposure corresponds to approximately 1.2 additional percentage points of deficit. This is modest relative to the five-point benchmark, yet it is large enough to suggest that

administrative borders can redistribute local access pressure even when the own-assignment average remains close to zero.

The estimate remains positive after several diagnostic changes. Controlling for total baseline cross-border reliance gives +0.250. Excluding the lowest-coverage HRR gives +0.222. Removing the largest outcome mover yields +0.163, while removing the three largest movers yields +0.159. The travel-time counterpart is +11.4 minutes per unit of exposure, which translates to about 0.67 minutes for a one-standard-deviation increase. Its randomization p-value is 0.116.

Exposure means are 0.0334 in both treatment arms under the realized assignment. This balance reduces concern that the measured exposure mechanically explains the average own-assignment contrast. It does not establish an absence of interference. The site-closure result also offers little support for a simple treated-neighbor exit channel. Facility-flow or station-level reallocation data would be needed to distinguish chain reallocation from other cross-border changes. The pattern is therefore most informative as evidence that the relevant market may extend beyond the randomized administrative unit.

B. BASELINE PAYMENT-PRESSURE PROXY

The second exploratory analysis asks whether the average result masks stronger responses in HRRs that entered ETC farther from its rewarded service mix. Facility-level payment adjustments are unavailable. We therefore construct a baseline proxy equal to one minus the average share of facilities offering peritoneal dialysis and home-hemodialysis training. A larger value indicates a larger baseline gap from the two home-dialysis services observed in the facility files.

The treatment interaction is +0.148 with an HC2 standard error of 0.067 and a randomization p-value of 0.018. A one-unit increase in the proxy therefore corresponds to 14.8 additional percentage points of deficit under treatment, though the observed proxy occupies a much narrower range. The coefficient should consequently be interpreted over realistic

Table 7: Treatment interacted with a baseline payment-pressure proxy

Specification	Coefficient	SE	p
Treatment × proxy (mean of the two gaps)	+0.148	0.067	0.018
with the rural interaction included	+0.178	0.071	
excluding the lowest-coverage region	+0.181		0.0007
excluding the largest mover	+0.123		0.036
excluding the three largest movers	+0.150		0.0013
Treatment × peritoneal-dialysis gap only	+0.100	0.058	0.092
Treatment × home-hemodialysis-training gap only	+0.137	0.066	0.021
Both interactions jointly (Wald test, 2 df)			0.054

Notes: Rows report coefficients on treatment interacted with the centered proxy. Regressions include census-region strata, treatment, the centered baseline outcome, its interaction with treatment, the proxy, and the treatment interaction. Standard errors are HC2. Randomization p-values use 20,000 stratified draws. The proxy equals one minus the mean baseline shares of facilities offering peritoneal dialysis and home-hemodialysis training. Component rows use each service gap separately. The joint row tests both interactions with a two-degree-of-freedom statistic.

differences in the proxy rather than as a literal 14.8-point forecast. The estimate is +0.181 after excluding the lowest-coverage HRR. It is +0.123 after removing the largest outcome mover and +0.150 after removing the three largest movers.

The proxy is nearly uncorrelated with rural share and baseline deficit. Adding the rural interaction gives a treatment-by-proxy coefficient of +0.178. The component estimates are less uniform. The peritoneal-dialysis gap gives +0.100 with a randomization p-value of 0.092. The home-hemodialysis-training gap gives +0.137 with a p-value of 0.021. A model containing both interactions has a joint randomization p-value of 0.054. The concentration in the training component suggests a specific service-capability channel, while the mixed component evidence also limits the strength of that interpretation.

Among the 101 HRRs in the top proxy tercile, 28 are treated. Their raw change-score contrast is +3.0 percentage points. This descriptive estimate points toward deterioration but remains inside the five-point materiality band. The proxy does not measure realized facility-level payment exposure and may capture other features of local supply. Its main use is to identify a margin for future work with actual payment adjustments.

VII. Interpretation and Scope

A. POLICY INTERPRETATION

The ETC model exposed a nationwide dialysis network to mandatory payment risk for almost five years. Across the measured facility configuration, the average network response is small. The primary confidence interval excludes an increase of five percentage points in the population without a second site inside 30 minutes. Randomization-test inversion supports a narrower ±2.5-point equivalence frontier under the constant-additive model. Continuous travel, closures, station capacity, and service availability provide no evidence of a compensating response that the threshold outcome missed. The result therefore concerns network resilience directly: the payment model did not create an average contraction large enough to materially reduce the availability of a backup in-center site.

This result resolves a supply-side question left open by the ETC evaluations. Patient-level studies show little average change in home dialysis or transplantation (Koukounas et al., 2024a; Dixit et al., 2026). The present evidence shows little average change in the physical network or in the home-dialysis services reported by facilities. Taken together, the model produced limited reorganization on both its intended patient margins and the measured provider-supply margins during the operating period. The absence of a supply contraction also clarifies the patient-side null: weak changes in modality use were not offset by a hidden average loss of physical access in the public facility data.

The bounded null also informs the economics of provider adjustment. Downside payments reached a meaningful share of treated facilities, yet the national network absorbed them without detectable average retrenchment. This pattern is consistent with substantial adjustment costs in a concentrated, fixed-cost industry. Chains may have absorbed modest losses within broader portfolios, and facilities may have adjusted on unobserved margins such as staffing or scheduling. The model's early termination also shortened the horizon over which exit could occur. The available data cannot separate these explanations, but each implies that physical

network reorganization is slower or less sensitive than the payment mechanism alone might suggest.

The cross-border pattern identifies a broader issue for place-based payment design. Administrative assignment occurs at HRR boundaries, while patients and multi-market chains operate across them. Average regional contrasts can remain near zero even when exposure shifts across neighboring markets. This matters for policy design because a region can export part of its adjustment burden to patients whose nearest alternatives lie across the border. Future evaluations can address this problem directly by linking assignment to facility flows and realized payment adjustments.

B. SCOPE AND LIMITATIONS

The outcomes describe facility configuration and potential geographic access in the geocoded CMS universe. They cover physical sites, dialysis stations, selected service flags, and road-time redundancy. They do not observe realized patient travel or facility choice (Salerno et al., 2022). Staffing and congestion remain outside the data, as do network acceptance and clinical quality.

The measurement universe includes 91.25 percent of eligible facility rows. Exact geocoding rates are similar across treatment arms, but the estimand applies to this geocoded universe. Routing is complete for 99.76 percent of CKD-risk mass. Worst-case bounds address the unresolved 0.24 percent for the threshold outcomes. Two HRRs have especially low routing coverage, and excluding both leaves the primary conclusion unchanged.

The endpoint captures month 59 of a model terminated early. The July 2026 vintage supplies a six-month post-termination check. Longer-run exits after payment adjustments ended remain outside the design. The March 2025 termination announcement may also have changed late-period expectations, although the available trajectory shows no discrete break.

Equivalence applies to average assignment effects within the stated band. The secondary interaction tests have limited power, so smaller subgroup-specific or HRR-specific harms

remain possible. The Wald confidence intervals use a no-interference variance construction. The randomization TOST uses a constant-additive-effect model. The global sharp-null randomization tests require neither of those specific restrictions, but they answer a different hypothesis.

VIII. Conclusion

Did mandatory payment risk reorganize the geography of dialysis supply? The CMS lottery and a fixed-network access measure show no material average change during the ETC model. At the 2025 endpoint, assignment changes the share of CKD-risk-weighted population without a second in-center site inside 30 minutes by -0.64 percentage points. The 90 percent confidence interval lies inside the ± 5-point band, while randomization-test inversion supports equivalence down to ± 2.5 points under its constant-effect model. The central finding is a quantitative bound on average deterioration. The confidence interval and randomization-test inversion define the magnitude of deterioration that the design can exclude.

The same conclusion appears in direct measures of travel, facility exit, station capacity, and home-dialysis service availability. The exploratory estimates at HRR borders and along the baseline payment-pressure proxy are smaller than the materiality band and require new data for confirmation. They nevertheless indicate where an average regional null may conceal localized pressure. Realized facility-level adjustments would allow a direct dose-response analysis. Patient claims and facility-flow records could then connect network changes to actual travel, switching, and hospitalization.

The policy implication follows from the bound on average network contraction. ETC did not achieve a detectable average reorganization of measured dialysis supply before its early termination, and it did not produce the average geographic contraction that critics feared. Local disruptions can still matter, especially where access crosses administrative borders. Future mandatory payment models can use the same geographic accounting to evaluate provider resilience alongside the patient outcomes the incentives are designed to change.

Data Availability

The study uses publicly available CMS facility archives and the published ETC assignment report, together with public geographic and population sources described in Appendix B. The source agencies, construction rules, and geographic inputs used to build the analysis are documented in that appendix.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used OpenAI's ChatGPT in order to assist with writing, language editing, and LaTeX formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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A. Supplementary Tables and Figures

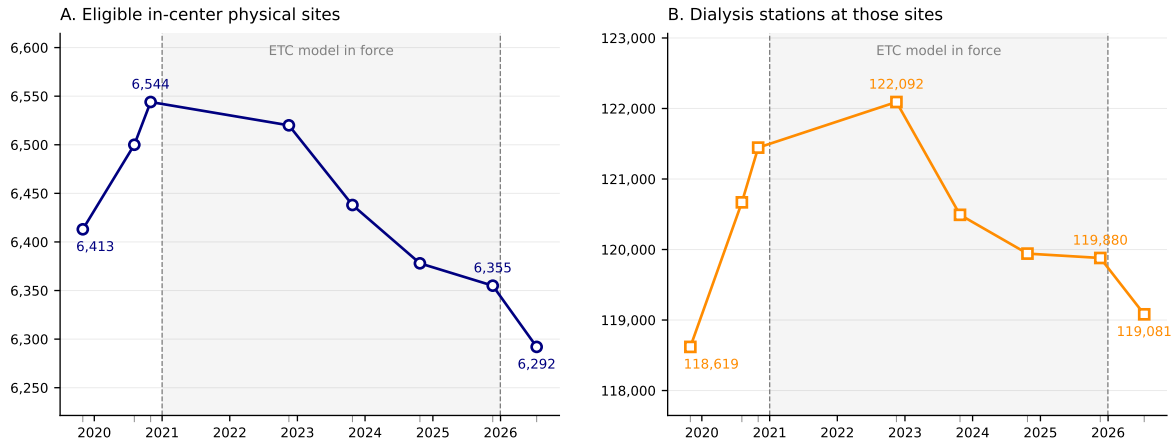


Figure A1: National counts of eligible in-center physical sites and dialysis stations by facility vintage. Vertical dashed lines mark ETC implementation and termination.

Table A1: Endpoint contrast across travel-time thresholds

Threshold (min)	Baseline mean (pct)	Estimate	SE	95% CI	Simultaneous 95% band	p
10	65.9	+0.44	0.77	[−1.07, +1.95]	[−1.52, +2.40]	
15	50.1	+0.21	0.87	[−1.51, +1.92]	[−2.02, +2.43]	0.82
20	41.0	−0.14	0.89	[−1.88, +1.60]	[−2.40, +2.13]	0.88
25	34.3	−0.38	0.94	[−2.22, +1.46]	[−2.77, +2.01]	0.69
27	31.9	−0.68	0.96	[−2.56, +1.20]	[−3.12, +1.76]	0.49
30	28.6	−0.64	0.95	[−2.50, +1.21]	[−3.05, +1.77]	0.52
33	25.3	−0.86	0.99	[−2.81, +1.08]	[−3.39, +1.66]	0.41
35	23.2	−0.87	1.01	[−2.86, +1.12]	[−3.46, +1.72]	0.42
40	18.8	−0.75	0.99	[−2.70, +1.19]	[−3.28, +1.78]	0.48
45	14.8	−0.72	0.87	[−2.42, +0.98]	[−2.93, +1.49]	0.43
50	11.8	−0.54	0.81	[−2.13, +1.04]	[−2.60, +1.52]	0.54
55	9.5	−0.41	0.78	[−1.93, +1.12]	[−2.39, +1.58]	
60	7.8	−0.26	0.79	[−1.81, +1.28]	[−2.27, +1.74]	0.77

Notes: Adjusted estimates are in percentage points at selected thresholds, with stratified Neyman standard errors and pointwise 95 percent confidence intervals. The simultaneous band uses the sup-t critical value from 5,000 joint sharp-null randomization draws over all 51 thresholds. Baseline means are equal-HRR deficit shares. Named-threshold p-values use 20,000 stratified draws.

B. Data Construction

Figure A2 summarizes the construction sequence. Facility rows enter when they contain a valid six-digit CCN, are located in the 50 states or the District of Columbia, and report

Table A2: Design precision as a function of baseline-adjustment variance reduction

Variance reduction from adjustment	SE (pp)	MDE at 80% power (pp)
0.000	3.26	9.14
0.500	2.31	6.46
0.700	1.79	5.00
0.800	1.46	4.09
0.900	1.03	2.89
0.916	0.95	2.65
0.950	0.73	2.04

Notes: The unadjusted stratified contrast has a standard error of 3.26 percentage points. Each row scales that standard error by the square root of one minus the stated variance reduction. The realized reduction implied by the adjusted and unadjusted standard errors is 0.916, while the endpoint-on-baseline R^2 is 0.891. Minimum detectable effects use a two-sided 5 percent test with 80 percent power.

in-center hemodialysis (Centers for Medicare & Medicaid Services, 2026). Addresses are canonicalized under a fixed rule and sent to the Census Bureau batch geocoder (U.S. Census Bureau, 2023). Rows with exact matches form the measurement universe.

We link rows into physical sites through canonical address equality and spatial continuity. Two rows within 100 meters are linked when at least two of the facility name, CCN, certification date, or chain fields agree. Pairs between 100 and 250 meters receive treatment-blind adjudication. Each linked site retains a single HRR assignment across vintages.

Origins are adult-population-weighted centroids of populated 2010 block groups (U.S. Census Bureau, 2012). We multiply 2014–2018 ACS adult population by 2020 PLACES tract-level CKD prevalence (U.S. Census Bureau, 2019; Centers for Disease Control and Prevention, 2020). A block group receives zero analysis weight when its tract lacks a PLACES estimate or when no comparable ACS geography exists. Point-in-polygon assignment maps origins and sites to Dartmouth Atlas HRRs (Wennberg and Cooper, 1996). A nearest-polygon rule handles coastal and island points outside all polygons.

The July 2019 OpenStreetMap planet state is reconstructed from the official full-history file (OpenStreetMap contributors, 2019) and compiled with OSRM backend 26.5.0 under the car profile (Luxen and Vetter, 2011; Project OSRM, 2026). For each origin and vintage, table queries return travel time to every candidate open site through a lossless candidate index.

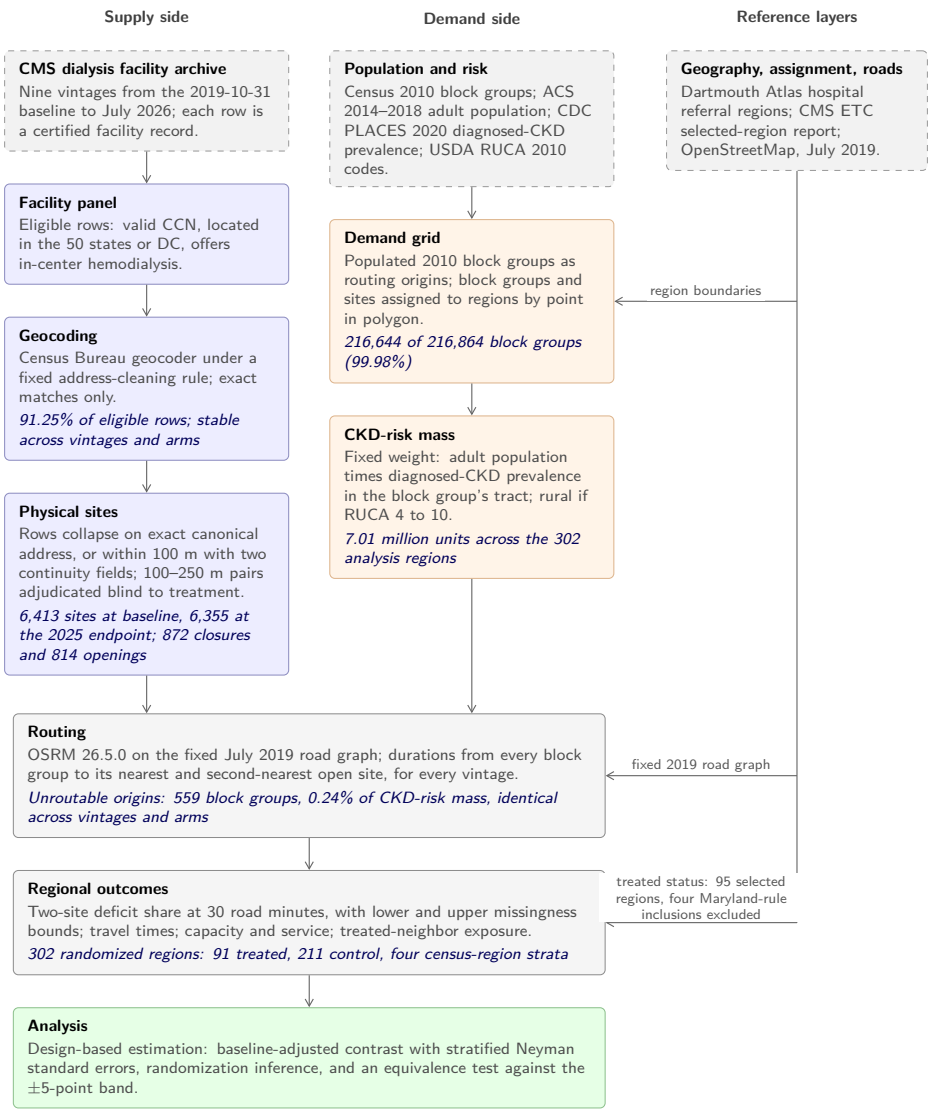


Figure A2: Data construction from public sources to regional analysis outcomes.

The two smallest resolved times define nearest-site and second-site access.

An origin is routing-complete when every candidate route required by the index is resolved. An origin whose nearest two candidates have no road path enters the deficit state at every threshold. Origins with unresolved candidates enter the missing-mass bounds in equation (2). The exposure matrix records, for each ordered HRR pair, the share of the origin HRR's baseline first-two access slots supplied by sites in the destination HRR. Capacity outcomes come from the linked site ledger, while service indicators come from the raw CMS facility files mapped through that ledger.